

Hindalco Summer Training Project

Project Report File

2026

Ai Predictive Energy Monitoring System



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➤ Abstract

Energy management has become a major concern for industries due to increasing energy costs, environmental challenges, and the need for sustainable operations. Efficient monitoring and analysis of energy consumption can help organizations optimize resource utilization, reduce operational expenses, and improve overall productivity. However, traditional methods of recording and analysing energy usage are often manual, time-consuming, and prone to errors.

The **AI-Based Energy Consumption Monitoring and Analysis System** is designed to provide a centralized platform for monitoring, storing, analysing, and predicting energy consumption across different departments within an organization. The system enables administrators to record daily energy usage, maintain historical records, and generate analytical reports based on daily, weekly, and monthly consumption patterns. An interactive dashboard provides graphical visualization of energy trends, department-wise comparisons, and overall energy statistics to support informed decision-making.

The project is developed using **React.js** for the frontend, **Node.js** and **Express.js** for the backend, and **MongoDB** as the database. A **Linear Regression** machine learning model, implemented in **Python** using the **Scikit-learn** library, is used to analyse historical energy consumption data and predict future energy requirements. This predictive capability helps management identify potential increases in energy demand and implement strategies for efficient energy planning.

The proposed system offers several advantages, including centralized data management, real-time monitoring, improved reporting accuracy, enhanced visualization through interactive charts, and AI-based forecasting of energy consumption. It reduces manual effort, supports data-driven decision-making, and contributes to better energy utilization and cost optimization.

The project demonstrates how the integration of web technologies, database management, data visualization, and artificial intelligence can provide an effective solution for industrial energy monitoring. Furthermore, the system can be enhanced in the future by integrating IoT-based smart energy meters, cloud computing, real-time sensor data, mobile applications, and advanced machine learning algorithms for more accurate predictions and automated energy optimization.

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➤ Introduction

Background

Energy plays a vital role in industrial operations. Efficient energy management helps reduce operational costs, improve productivity, and support sustainable development. Different departments consume different amounts of electricity depending on their activities and equipment. Therefore, monitoring and analysing energy consumption is essential for effective resource utilization.

Traditional methods of recording energy consumption are mostly manual, making them time-consuming and prone to errors. An automated system can simplify data management, improve accuracy, and provide valuable insights for decision-making.

Project Introduction

The **AI-Based Energy Consumption Monitoring and Analysis System** is a web-based application developed to monitor, store, analyse, and predict department-wise energy consumption. The system records daily energy usage, maintains historical data, and provides graphical reports through an interactive dashboard.

A **Linear Regression** machine learning model is used to analyse historical energy data and predict future energy consumption, helping organizations plan and optimize energy usage.

Problem Statement

Many industries do not have a centralized system to monitor energy consumption across different departments. Manual record-keeping makes it difficult to analyse energy trends, identify high-consuming departments, and predict future energy requirements. This project provides an intelligent solution to overcome these challenges.

Objectives

The main objectives of the project are:

- To monitor department-wise energy consumption.
 - To maintain energy records in a centralized database.
 - To analyse daily, weekly, and monthly energy usage.
 - To generate graphical reports and dashboards.
 - To predict future energy consumption using Linear Regression.
 - To improve energy management and support better decision-making.
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Scope of the Project

The system allows users to manage departments, record daily energy consumption, view analytical reports, and predict future energy usage. It is designed for industrial organizations and can be enhanced in the future with IoT sensors and real-time monitoring.

Key Features

- Department Management
 - Energy Consumption Monitoring
 - Interactive Dashboard
 - Energy Usage Analysis
 - AI-Based Prediction
 - Report Generation
 - Secure Data Storage
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➤ Literature Review

Introduction

Energy management is an important aspect of industrial operations. Many industries use digital systems to monitor and analyse energy consumption to improve efficiency and reduce operational costs. With the advancement of Artificial Intelligence (AI) and Machine Learning (ML), energy prediction has become more accurate and effective.

Existing System

Traditional energy monitoring systems mainly depend on manual data entry or spreadsheets. These systems are time-consuming, prone to errors, and do not provide intelligent prediction or advanced analysis.

Proposed System

The proposed AI-Based Energy Consumption Monitoring and Analysis System provides a centralized platform to record, analyse, and predict energy consumption. It uses **React.js**, **Node.js**, **MongoDB**, and a **Linear Regression** model to generate reports, visualize energy trends, and predict future energy consumption.

➤ System Analysis

Introduction

System analysis is the process of understanding the requirements of a system and designing a solution that meets user needs. The AI-Based Energy Consumption Monitoring and Analysis System is developed to provide an efficient way to monitor, analyse, and predict energy consumption.

Functional Requirements

The system provides the following functions:

- Manage department details (Add, Update, Delete, View).
- Record daily energy consumption.
- Store energy records in the database.
- Display energy consumption reports and charts.
- Predict future energy consumption using Linear Regression.
- Generate analysis reports for decision-making.

Non-Functional Requirements

The system should:

- Provide a user-friendly interface.
- Ensure secure and reliable data storage.
- Deliver fast response and efficient performance.
- Support scalability for future enhancements.
- Maintain data accuracy and availability.

Feasibility Study

Technical Feasibility

The project uses React.js, Node.js, MongoDB, and Python, which are open-source and widely used technologies.

Economic Feasibility

The system is cost-effective because it uses free development tools and open-source software.

Operational Feasibility

The application is easy to use and helps organizations efficiently monitor and manage energy consumption.



System Architecture

Introduction

The AI-Based Energy Consumption Monitoring and Analysis System follows a **three-tier architecture**, consisting of the Presentation Layer, Application Layer, and Database Layer. The system allows users to record energy consumption, analyse data, and predict future energy usage through a web-based interface.

System Architecture

The architecture consists of the following components:

- **Presentation Layer (Frontend):**
 - Developed using React.js.

- Provides a user-friendly interface for managing departments, recording energy data, viewing reports, and displaying charts.
 - **Application Layer (Backend):**
 - Developed using Node.js and Express.js.
 - Processes user requests, performs business logic, and communicates with the database and AI model.
 - **Database Layer:**
 - MongoDB stores department details, energy consumption records, and prediction data securely.
 - **Machine Learning Layer:**
 - Python with Linear Regression analyses historical energy data and predicts future energy consumption.
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Architecture Flow

1. The user enters energy consumption data through the React.js frontend.
 2. The frontend sends the request to the Node.js and Express.js backend.
 3. The backend stores and retrieves data from MongoDB.
 4. Historical data is sent to the Python Linear Regression model for prediction.
 5. The predicted results are returned to the backend.
 6. The dashboard displays reports, charts, and prediction results to the user.
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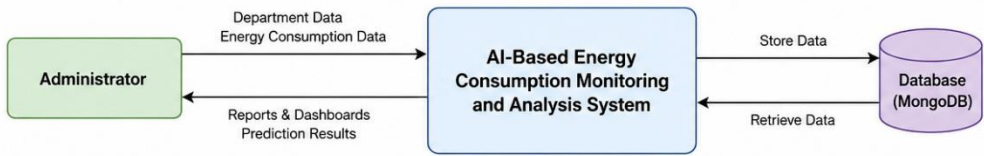


Data Flow Diagram (DFD)

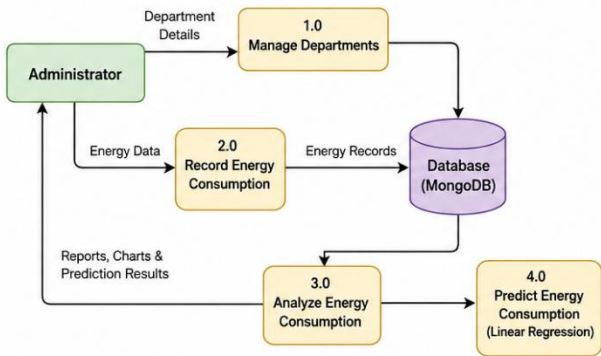
Introduction

A Data Flow Diagram (DFD) is a graphical representation of how data moves through the system. It illustrates the flow of information between users, processes, and the database. The AI-Based Energy Consumption Monitoring and Analysis System uses DFDs to describe the overall workflow of data collection, storage, analysis, and prediction.

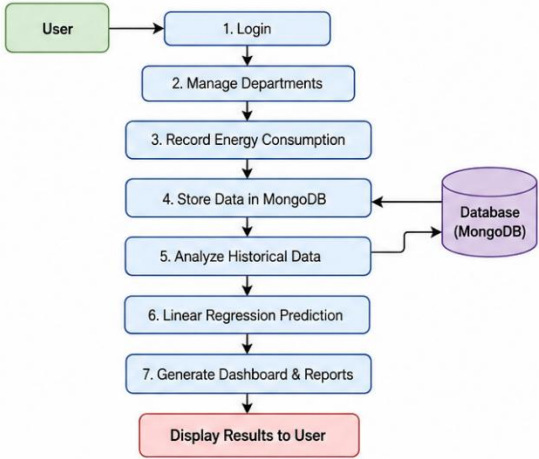
DFD Level 0 (Context Diagram)



DFD Level 1

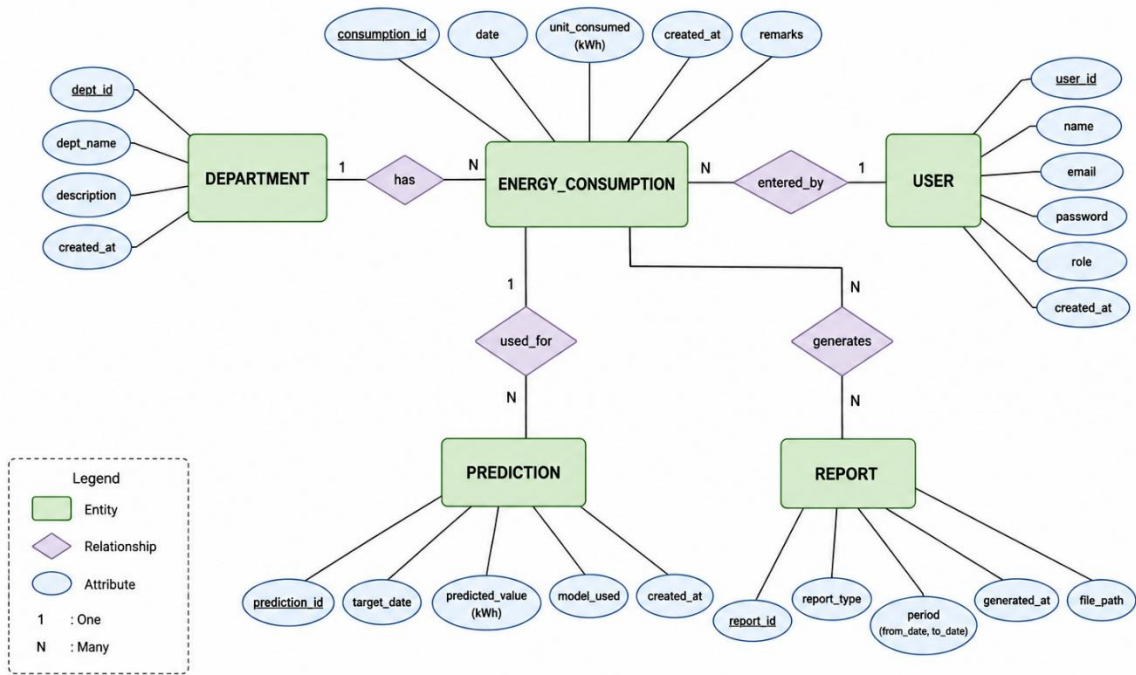


DFD Level 2



ER Diagram

ER DIAGRAM





Database Design

Introduction

The database is designed to store department details, energy consumption records, and prediction data in a structured manner. **MongoDB** is used as the database because it is flexible, scalable, and suitable for handling large amounts of data.

Department Collection

This collection stores information about different departments.

Field Name	Data Type	Description
Departmentwide	String	Unique department ID (Primary Key)
Department Name	String	Name of the department
Location	String	Department location

Energy Consumption Collection

This collection stores daily energy consumption records for each department.

Field Name	Data Type	Description
EnergidID	String	Unique energy record ID (Primary Key)
DepartmentwideID	String	Reference to Department ID (Foreign Key)
EnergyConsumedID	Number	Energy consumed in kWh
Date	Date	Date of energy consumption

Prediction Collection

This collection stores the predicted energy consumption generated by the AI model.

Field Name	Data Type	Description
PredictionID	String	Unique prediction ID (Primary Key)
DepartmentID	String	Reference to Department ID (Foreign Key)
PredictedConsumption	Number	Predicted energy consumption (kWh)

Field Name	Data Type	Description
PredictionDate	Date	Date of prediction



API Documentation

Introduction

The AI-Based Energy Consumption Monitoring and Analysis System uses **RESTful APIs** to enable communication between the React.js frontend, the Node.js/Express.js backend, and the MongoDB database. These APIs allow users to perform CRUD operations and access analysis and prediction features.

API Endpoints

Department APIs

Method	Endpoint	Description
GET	/api/departments	Retrieve all departments
POST	/api/departments	Add a new department
PUT	/Api/departments/:id	Update department details
DELETE	/Api/departments/:id	Delete a department

Energy Consumption APIs

Method	Endpoint	Description
GET	/api/energy	Retrieve all energy records
POST	/api/energy	Add a new energy record
PUT	/Api/energy/:id	Update an energy record
DELETE	/Api/energy/:id	Delete an energy record

Prediction API

Method	Endpoint	Description
POST	/api/predict	Predict future energy consumption using the AI model



Key Features

The AI-Based Energy Consumption Monitoring and Analysis System provides the following key features:

- **Department Management:** Add, update, delete, and view department details.
- **Energy Monitoring:** Record and maintain daily department-wise energy consumption.
- **Centralized Database:** Securely store historical energy consumption data using MongoDB.
- **Data Analysis:** Analyse daily, weekly, and monthly energy consumption trends.
- **Interactive Dashboard:** Display energy statistics and graphical reports using charts.
- **AI-Based Prediction:** Predict future energy consumption using the Linear Regression algorithm.
- **Report Generation:** Generate department-wise and monthly energy consumption reports.
- **User-Friendly Interface:** Easy-to-use web application developed with React.js.
- **Data Accuracy:** Maintain organized and reliable records for better decision-making.
- **Scalability:** The system can be enhanced with IoT sensors, real-time monitoring, and advanced AI models in the future.



Benefits

The AI-Based Energy Consumption Monitoring and Analysis System offers the following benefits:

- Improves energy monitoring through a centralized management system.
- Reduces manual effort and minimizes human errors in data recording.
- Stores historical energy consumption data securely in a MongoDB database.
- Provides graphical reports and dashboards for easy analysis.
- Identifies departments with high energy consumption.
- Predicts future energy usage using the Linear Regression algorithm.
- Supports better planning and data-driven decision-making.
- Helps reduce energy wastage and operational costs.
- Increases efficiency by automating energy monitoring and reporting.
- Can be easily upgraded with IoT devices and advanced AI models for real-time monitoring.



Implementation

Introduction

- The AI-Based Energy Consumption Monitoring and Analysis System is implemented using a three-tier architecture consisting of the frontend, backend, database, and machine learning model. The system is designed to provide efficient energy monitoring, analysis, and prediction through a web-based interface.
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Frontend Implementation

- The frontend is developed using **React.js**. It provides a user-friendly interface for managing departments, recording energy consumption, viewing reports, and displaying dashboards with graphical charts.
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Backend Implementation

- The backend is developed using **Node.js** and **Express.js**. It handles user requests, processes business logic, manages REST APIs, and communicates with the MongoDB database and the AI prediction model.
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Database Implementation

- **MongoDB** is used to store department details, energy consumption records, and prediction data. The database ensures secure and efficient data storage and retrieval.
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AI Model Implementation

- The prediction module is developed in **Python** using the **Scikit-learn** library. A **Linear Regression** model is trained on historical energy consumption data to predict future energy usage.

➤ Challenges Faced

During the development of the AI-Based Energy Consumption Monitoring and Analysis System, several challenges were encountered:

- Collecting and organizing accurate historical energy consumption data.
- Designing a user-friendly and responsive interface.
- Integrating the React.js frontend with the Node.js and Express.js backend.
- Establishing a reliable connection with the MongoDB database.
- Implementing and training the Linear Regression model for prediction.
- Ensuring accurate data analysis and visualization using charts.
- Managing API communication between different system components.
- Testing the system to ensure data accuracy and smooth performance.

Despite these challenges, the project was successfully implemented by using modern web technologies and machine learning techniques, resulting in an efficient and reliable energy monitoring system.

➤ Conclusion

The **AI-Based Energy Consumption Monitoring and Analysis System** provides an efficient and centralized solution for monitoring, analysing, and predicting energy consumption across different departments. The system simplifies the process of recording energy usage, maintaining historical data, and generating analytical reports through an interactive dashboard.

By integrating **React.js**, **Node.js**, **Express.js**, **MongoDB**, and a **Linear Regression** machine learning model, the project enables organizations to make informed decisions based on accurate energy consumption analysis and future predictions. It helps reduce manual effort, improve data accuracy, and support better energy management.

Overall, the proposed system is reliable, user-friendly, and scalable. It can be further enhanced by integrating IoT devices, real-time data monitoring, and advanced machine learning techniques to achieve smarter and more efficient energy management.

➤ Screenshots-Project

